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IT4010 Research Project – 2026 Jan

Topic Assessment Form

V2.1

Project ID: R26-IT-112

1. Topic (12 words max)

A Next-Generation AI Framework for Intelligent Travel Support in Sri Lanka

2. Research group the project belongs to

SST - Software Systems & Technologies

3. Specialization of the project belongs to

Information Technology (IT)

4. If a continuation of a previous project:

Project ID	No
Year	No

5. Brief description of the research problem including references (300 words max) – references not included in word count.

Sri Lanka's tourism industry, a cornerstone of the economy contributing 4-5% to GDP pre-crisis, confronts persistent infrastructure deficits, climate vulnerabilities, and planning inefficiencies that undermine sustainable growth and visitor satisfaction [1]. Following the 2022 economic crisis and the 2025 shortfall of 500,000 tourists against the 3 million target (achieving only 2.5 million arrivals with US$4 billion revenue), critical challenges endure: inadequate transport connectivity, exemplified by non-linear road/rail travel times (100km journeys spanning 1-5 hours) delays access to iconic UNESCO sites like Sigiriya Rock Fortress and Sinharaja Rainforest. Monsoon-induced landslides and rising sea levels disrupt peak seasons, while fragmented information ecosystems overwhelm international visitors (primarily from India, China, Europe) with language barriers, rigid generic itineraries, and absent real-time intelligence on dynamic factors like weather, crowd densities at Yala National Park, or nationality-specific Electronic Travel Authorization (ETA) compliance.

Regional disparities compound issues, with Colombo and coastal hubs facing overcapacity while secondary destinations like Ella and Mirissa suffer chronic low occupancy (under 40%). Personalization deficits fail diverse segments from budget backpackers seeking eco-treks to luxury travelers desiring wellness retreats resulting in suboptimal satisfaction scores and uneven economic spillovers [2]**]**. Existing digital tools, including static apps (TripAdvisor, Google Maps) and nascent local chatbots, deliver reactive, siloed functionalities (e.g., basic bookings) but neglect holistic end-to-end orchestration. They overlook Sri Lanka-unique constraints, variable terrain speeds, cultural sensitivities, emotion-aware adaptations via multi-modal inputs and proactive safety protocols amid data silos and digital infrastructure gaps.

These gaps reveal a pressing research void: no integrated frameworks exist fusing Large Language Models (LLMs) with live IoT feeds (weather APIs, transport sensors) for agentic, multi-modal (voice/text) travel guardianship ensuring compliance, safety, and sustainability [3] This project pioneers such a next-generation AI system to deliver hyper-personalized planning, real-time adaptive re-routing, and analytics-driven insights, fostering resilient sector transformation.

References

[1] S. L. T. Alliance, 2025.

[2] PLN4, "How AI is Revolutionizing Sri Lankan Tourism," 2025.

[3] L. N. Web, "Sri Lanka Unveils First AI-Driven Hotel in Landmark Tourism Milestone," October 2025.

6. Brief description of Existing Research and Systems (300 words max)

Research on smart travel planning has come a long way from basic recommendation setups to stuff involving large language models and optimization for multiple goals. Traditional sites like TripAdvisor just give out fixed info, but generative AI can make personalized itineraries that fit people better, as Arachchi pointed out in 2022 [1]. Still, the Technology Acceptance Model suggests users do not stick with systems that ignore real life issues, like those ground truths in logistics. It feels like theres a big hole in building agent systems, something like the Tri-Agent engine that does more than chatbots by adjusting routes proactively with voice, especially for tricky spots in Sri Lanka, according to Mustafa back in 2019 [2].

When it comes to safety in transportation, apps such as PickMe and Uber stick to GPS and driver ratings mostly. Academic work looks at better ways, like biometrics for checking identities and IoT for watching things closely, so both drivers and passengers get verified on both ends. But there is not really an AI model that scores risks all the time, keeping an eye on how someone drives or if they follow the route. Existing systems focus on transactions, and they do not connect tourists, drivers, and owners into one safety net that works fully.

Affective computing works well in lab settings, but not so much in actual tourism yet. For example, studies in IEEE Access from 2024 show 3D convolutional neural networks with attention can pick up emotions like anxiety in emergencies, getting over 88 percent accuracy [3]. These are stuck in research though, not turned into adaptive decision support systems. Destination management does not track emotions over time or make heatmaps for engagement right away. This project tries to fix that by making emotion recognition active, helping optimize places and keep tourists safer.

Immigration and compliance stuff is mostly handled through boring government sites or visa helpers that do not change. Weerakoon in 2024 says the problem is no real link between rules and planning tools, which blocks smart tourism from happening [4] Rule based systems show up in legal tech, but they are not plugged into tourism as helpers for decisions. So the literature points to needing automated checks that look at where someone is from and their plans in real time, to stop bad travel ideas before they lock in, making immigration more proactive instead of just manual. It seems like that shift could make a difference, though I am not sure how easy it would be to pull off.

References

[1] P. L. L. C. X. Z. M. L. F. L. Yiming Du, "Attention-based 3D convolutional recurrent neural network model for multimodal emotion recognition," *Frontiers in Neuroscience,* p. 7516–7521, 10 January 2024.

[2] A. M. M. Mustafa, "The Impact of Infrastructure on Tourism: The Case of Sri Lanka," *Asian Social Science,* p. 174–174, 30 June 2019.

[3] L. W. X. J. Ben Zhou, "Multi-Scale Convolutional Attention and Riemannian Geometry Network for EEG-Based Motor Imagery Classification," *IEEE Access,* January 2024.

[4] D. Dharshana Weerakoon, "Digital Innovation in Sustainable Wellness: A Strategic Framework for Transforming Sri Lanka's Tourism and Hospitality Landscape," Institute of Professional Studies & Skill Development / Sustainable Wellness Group Ltd, London / Sri Lanka, 2025.

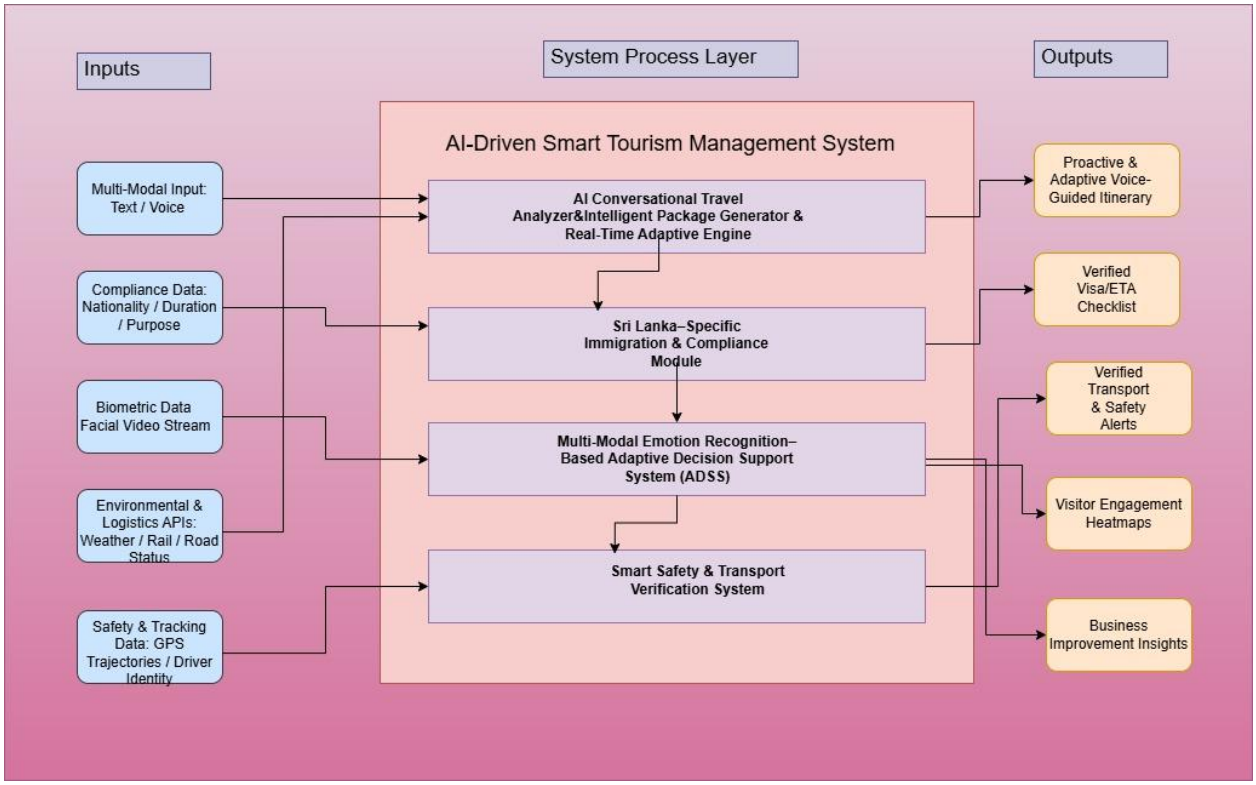
7. Brief description of the solution's nature, including a conceptual diagram (maximum 500 words).

The first component, the **AI Conversational Travel Analyzer, Intelligent Package Generator, and Real-Time Adaptive Engine**, solves the dual problems of fragmented planning and reactive service. By utilizing a Tri-Agent architecture, it transforms messy, multi-intent user inputs into structured itineraries using a Genetic Algorithm-based optimizer tailored to Sri Lankan logistical constraints. Most importantly, it introduces a "Proactive Guardian" logic that monitors live environmental APIs to initiate voice-driven re-routing. This shifts the travel experience from a reactive model—where the tourist must solve problems—to a proactive agentic model that resolves disruptions before the tourist encounters them.

The **Sri Lanka-Specific Immigration and Compliance Module** addresses the critical gap of legal silos within the tourism journey. Currently, tourists must navigate the complex Electronic Travel Authorization (ETA) and visa protocols independently of their travel planning. This component provides a rule-based AI engine that integrates directly with the itinerary generator to verify the legal feasibility of a trip based on nationality and duration. By automating document checklists and providing real-time compliance alerts, it ensures that immigration management is a seamless, risk-free part of the digital ecosystem rather than a manual, error-prone hurdle.

The **Adaptive Emotion Recognition for Tourist Experience Management** addresses the lack of emotional intelligence in destination management. While existing systems provide static reviews, this component utilizes a Multi-Modal Adaptive Decision Support System (ADSS) to track "Emotional Trajectories" through facial landmarking and temporal analysis. By correlating emotions with environmental context (e.g., specific zones in a heritage site), the system can detect anomalies like prolonged distress or boredom. This allows authorities to optimize site layouts and provides the platform with the data needed to suggest real-time itinerary changes based on the user's actual satisfaction levels.

The **Smart Safety & Transport Verification System** provides a robust solution to the safety concerns and lack of transparency in local transportation. While current ride-sharing services offer basic tracking, this component introduces a dual Face-ID verification system for both tourists and drivers, ensuring authorized participation. It further employs an AI-based risk scoring model that monitors driving behavior and route adherence in real-time. By integrating vehicle owner notifications and automated emergency alerts for authorities, it creates a "verified safe" loop that protects the foreign traveler throughout their physical transit across the country.


Inputs
System Process Layer
Outputs
AI-Driven Smart Tourism Management System
Multi-Modal Input: Text / Voice
Compliance Data: Nationality / Duration / Purpose
Biometric Data Facial Video Stream
Environmental & Logistics APIs: Weather / Rail / Road Status
Safety & Tracking Data: GPS Trajectories / Driver Identity
AI Conversational Travel Analyzer&Intelligent Package Generator & Real-Time Adaptive Engine
Sri Lanka–Specific Immigration & Compliance Module
Multi-Modal Emotion Recognition–Based Adaptive Decision Support System (ADSS)
Smart Safety & Transport Verification System
Proactive & Adaptive Voice-Guided Itinerary
Verified Visa/ETA Checklist
Verified Transport & Safety Alerts
Visitor Engagement Heatmaps
Business Improvement Insights

8. Brief overview of the availability of necessary specialized domain expertise, knowledge, and data. (500 words max)

The project team will bridge the gap between technical AI development and the tourism industry by acquiring deep domain expertise through a multi-faceted approach. This involves a thorough analysis of the Sri Lanka Tourism Development Authority (SLTDA) strategic reports and the Department of Immigration and Emigration regulatory guidelines. To ensure the system handles local ground realities accurately, members will engage with local travel operators to understand specific logistical constraints, such as the seasonal impacts on the Ella-Kandy rail line and the cultural sensitivities required for site management in the Cultural Triangle. This domain knowledge is critical for tailoring the "Tri-Agent" system to respect Sri Lanka's unique social and geographical landscape.

Technical knowledge and skills will be developed through specialized training pathways tailored to each individual component. For the planning and immigration modules, members will undergo intensive training in Large Language Models (LLMs), Multi-objective Optimization (Genetic Algorithms), and Rule-Based AI logic to handle legal compliance. The modules focused on emotion-aware decision support and safety will require mastery of Computer Vision frameworks, specifically ResNet-18 for environmental recognition and Long Short-Term Memory (LSTM) networks for temporal emotional analysis. Furthermore, expertise in Biometric Authentication (Face-ID) and mobile sensor integration will be cultivated to engineer the real-time risk-scoring models necessary for the transport safety component.

Data collection will utilize a hybrid strategy of primary and secondary sources, supported by a rigorous ethical framework. Primary data will consist of simulated user interaction logs and facial landmark datasets used to train the emotion recognition module, alongside simulated GPS trajectories for common Sri Lankan routes. Secondary data will be fetched via REST APIs, including OpenWeather for climate alerts and Amadeus for aviation status. Given that the project processes biometric facial video data and human emotions, it strictly adheres to the Sri Lanka Personal Data Protection Act (No. 9 of 2022). The team will seek formal approval from the University's Ethics Review Committee (ERC), ensuring that the solution prioritizes data anonymization, informed consent via an opt-in mechanism, and high-level encryption for all biometric records.

9. Objectives and Novelty

Main Objective

To design and develop an AI-driven, integrated Smart Tourism Management System for Sri Lanka that provides personalized planning, immigration compliance, safety verification, and emotion-aware adaptive decision support to enhance the foreign tourist experience.

New Functionalities and Unique Features

- AI Conversational Travel Analyzer + Intelligent Package Generator + Real-Time Adaptive Engine
  Proactive "Guardian" Intervention: Unlike existing platforms (Google Maps, TripAdvisor) that only respond to user queries, this system introduces an event-driven Adaptive Engine. It monitors live data streams (weather, train delays, landslides) and proactively initiates Voice/SMS interventions to re-route tourists before they encounter disruptions.
- Sri Lanka–Specific Immigration & Compliance Module
  Integrated Legal-Planning Engine: The system introduces a first-of-its-kind synchronization between ETA (Immigration) compliance and travel planning. It ensures that itineraries are not just logically optimized but also legally feasible based on the tourist's nationality and visa regulations in real-time.
- Smart Safety & Transport Verification System
  Dual-Sided Safety Verification: The system introduces a Dual Face-ID Protocol that authenticates both the tourist and the driver. This creates a "Verified Safe" transport loop that automatically notifies vehicle owners of bookings and uses AI to generate real-time safety scores based on route adherence.
- Multi-Modal Emotion Recognition–Based Adaptive Decision Support System (ADSS)
  Emotion-Aware Site Management: We are introducing Multi-Modal Emotion Recognition that tracks "Emotional Trajectories." This allows the system to compute a Satisfaction Index (SI) and detect anomalies (stress/anxiety) in secure heritage sites, providing site managers with actionable engagement heatmaps.

Member Name with Registration No	Sub Objective	Tasks	Novelty
Student ID	Your component of the project	Tasks you perform to achieve your component and overall objectives.	Your genuine contribution / What you can't observe in other research projects that have tackled the same problem
IT22216014	Adaptive Emotion Recognition for Tourist Experience Management	• Monitor tourists' expressions and surroundings. • Detect unusual behavior indicating emergencies and provide suggestions for quick action or intervention. • Create real-time engagement maps. • Assess satisfaction across locations. • Identify low-satisfaction areas and provide recommendations to enhance visitor experience and improve business performance.	Combines face, environment, and time-based emotion analysis. Includes real-time feedback, anomaly detection, and location-specific improvement suggestions, which are not commonly found in other tourism management research. Transforms simple emotion detection into an intelligent management and decision-support tool that enhances tourist experiences at specific locations, identifies areas of low satisfaction, provides actionable recommendations to improve them, and supports business development by optimizing operations and increasing visitor satisfaction.
IT22176974	Smart Safety & Transport Verification System	• Implement Face-ID verification for both tourists and drivers to prevent	The Smart Safety & Transport Verification System is unique in its integration of multiple advanced

		unauthorized access or impersonation. • Develop a system to automatically notify vehicle owners when their vehicle is booked or reserved. Integrate GPS tracking to monitor real-time vehicle location, route adherence, and travel progress. Develop an AI-based risk scoring model to continuously evaluate transport safety based on driving behavior, route deviations, and other factors. • Implement real-time emergency alert notifications for tourists, vehicle owners, and authorities in case of incidents.	features that are not commonly found in existing research. It provides dual Face-ID verification for both the tourist and the driver, ensuring that only authorized individuals can use the service. The system automatically notifies vehicle owners in real time whenever their vehicle is booked or reserved, enhancing transparency and accountability. Continuous GPS tracking combined with AI-based safety monitoring allows for real-time evaluation of driving behavior, route adherence, and potential risks. In addition, the system can generate immediate emergency alerts to tourists, vehicle owners, and authorities in case of incidents. By combining verified transport authorization with AI-driven continuous safety monitoring and real-time alerts, this system offers an end-to-end solution that is innovative and not present in existing research.
IT22607614	AI-Based Immigration and Travel Compliance	• Analyze and formalize Sri Lanka visa and ETA regulations based on	Integrates Sri Lanka-specific immigration intelligence directly into AI-based tourism planning, which is

	Management for Foreign Tourists Visiting Sri Lanka	nationality, travel duration, and visit purpose. • Detect tourist nationality and travel details automatically from user profiles and itineraries. Design and implement a rule-based AI decision engine to evaluate visa/ETA eligibility. • Generate a personalized immigration compliance checklist including required documents and timelines. Integrate immigration validation with the AI travel planning engine to prevent invalid itineraries. • Provide automated alerts and reminders for visa expiry risks, missing documents, and compliance deadlines. Maintain a centralized database of immigration rules and tourist compliance records.	rarely addressed in existing tourism research. Unlike static visa information systems, this module acts as an active decision-support and risk-prevention engine that reasons over nationality, itinerary, and regulations in real time. The novelty lies in embedding immigration compliance before travel confirmation, ensuring only legally feasible travel plans are generated. This transforms immigration handling from a manual, error-prone process into an automated, intelligent, and proactive compliance mechanism, enhancing tourist convenience while supporting national tourism governance
IT22082442	AI Conversational Travel Analyzer + Intelligent Package Generator + Real-Time Adaptive Engine (Tri-Agent System)	• Develop a Multi-Intent NLP Analyzer: Using LLMs (e.g., Llama 3/GPT-4o) to extract semantic meaning, constraints, and user "vibes" from free-text inputs. • Implement an Intelligent Package Generator: Designing a Genetic	Shift from Reactive Support to Proactive Agentic Action: Unlike traditional research which focuses on reactive chatbots (user asks ->AI responds), this system is Agentic. It introduces a "Proactive Guardian" logic that identifies environmental

		Algorithm (GA) to solve the "Traveler Constraint Satisfaction Problem," optimizing for budget, time, and Sri Lanka's unique road/rail speeds. • Design the Real-Time Adaptive Engine: Building an event-driven "Guardian" loop that monitors live APIs (Weather, Train/Flight status) for disruptions. • Build a Proactive Voice Interface: Implementing a voice-driven notification system that initiates contact with the user to offer alternative plans during trip disruptions.	risks (e.g., a landslide in the Hill Country) before the traveler arrives and initiates a voice-guided re-routing. It specifically models Sri Lanka's non-linear travel constraints (where a 100km journey can take 1 hour on an expressway or 5 hours on a mountain road), a nuance missing in global generic planners.

10. <u>This part is to be filled by the Supervisor and the Co-supervisor of the Project.</u>

a) This research topic possesses a comprehensive scope suitable for a final-year project.

Yes	✓	No	

b) The proposed topic exhibits novelty.

Yes	✓	No	

c) The student group can successfully execute the proposed project.

Yes	✓	No	

d) The proposed sub-objectives reflect the students' areas of specialization.

Yes	✓	No	

e) Any other comments:

	Title	First Name	Last Name	Signature
Supervisor	Dr.	Kaushalya	Dissanayake	K. Dissanayake
Co-Supervisor	Ms.	Suriya	Kumari	
External Supervisor				
Summary of external supervisor's (if any) experience and expertise				